

local market opportunities and emerge as a leading cloud provider in India.

Q. Could you explain concepts like 'universal credits' and 'Bring your own Licence'?

One of the key things we realised, based on the feedback we got from our customers, is that they want flexibility with what they use, when they use it and how much of it they use. If you want them to commit to how much database capacity they want to buy, how much of integration capacity they want to invest in upfront, or how much of mobile cloud service they want to consume, then they actually don't know because utilisation in the cloud could suddenly scale up or not happen at all. Therefore, we introduced this notion of 'universal credits (UC)' pricing, whereby we give them the flexibility to buy a pool of credits from Oracle, with which they are entitled to use all our cloud, IaaS and PaaS services. Their usage is metered and adjusted against their universal credits. In addition, when we release a new software or new cloud service, they are entitled to use those too.

We are making it easier for a customer to be able to use the cloud without worrying how much they're using, what they're using, etc. This is what our universal credits are about. Think of it as a virtual prepaid wallet you can use to consume any Oracle platform/infrastructure cloud service, including mixing and matching, based on the quantum of universal credits you have opted for.

With 'Bring Your Own Licence (BYOL)', we want to enable portability from on-premise licences to the cloud. If customers decide to migrate to the cloud from their on-premise usage model, they don't have to pay double the cost for the licence. If they have licences that are unused (on-premises), they can actually bring that licence to the cloud, and there is no reason to charge them for the same licence again. Thus, the subscription price of that cloud service comes down significantly. This means they have to pay only for the infrastructure upon which it is running, and the automation and operations that we do. So that's the logic behind 'BYOL'.

Q. What is the role of Oracle's Indian centre with respect to development and other work?

Oracle India is home to our largest employee base outside of the US. Our teams in India are involved with all our product lines, and there's a lot of great collaboration happening. Oracle India is the only organisation outside Oracle's headquarters in California to represent all divisions including sales, marketing, consulting, as well as support and education operations for domestic and global clients. In fact, India accounts for the organisation's largest research and development investment outside US.

Q. With respect to developers, what is the USP of Oracle's cloud platform?

The developers can easily add intelligence to their platform. They can leverage a very open platform that can run in multiple environments, and not just within the public cloud. They can run several functions on-premises or within the public clouds. They're able to deliver hybrid capabilities to an application that can either work on-premises or on the public cloud. Our goal is to give them better functionality out-of-the-box — that is pre-assembled so that they can focus on their business differentiators and accelerate innovation.

At the end of the day, developers need rich and robust platform capabilities. They want the ability to choose any programming language, database, operating system, virtualisation technology and developer tool, including open source and third-party options. Oracle Cloud Platform includes key capabilities for embedded AI and machine learning to help developers build new AI based applications, and to enable IT and operations to automate the manual steps in provisioning, patching, tuning, and more. With the Oracle AI platform cloud service, developers and data scientists are empowered to develop and deploy machine learning models, and to manage the complete data science life cycle on their preferred open source frameworks.

Q. Coming to the Internet of Things (IoT) - are there platforms specifically designed for IoT applications?

A few years ago, we started a platform on which one could connect the gateways, gather the data coming from different sensors and make intelligence decisions at the edge. We can now aggregate the data into the public cloud, do predictive analytics on that data, and connect the devices to the back-end applications.

After building this platform, we realised that when you look at the usage of IoT, most people (for example, a person on the manufacturing plant floor or someone who is responsible for the servicing of goods) are not interested in building an application on the platform. They just want to use IoT and get the value out of it. Therefore, we built a set of SaaS applications on top of this platform. This platform is still available for anyone who wants to build an IoT app on it.

As of now, we have released a set of four SaaS applications as part of our IoT focus.

- **Asset monitoring:** This gives users the ability to monitor the assets, create a geofence, take sensor data from these assets, and get health information.
- **Production line monitoring:** When you are on a manufacturing floor, you can use this app to get insights from various sensors — the yield of a production line, and if the yield is going down in one line then how you can make it reconfigure itself, etc.
- **Fleet monitoring:** When there is a distribution of vehicles, this app helps you monitor these vehicles — how to monitor the drivers' behaviour, how you will figure out where the vehicles are, etc.
- **Connected workers:** This gives users the ability to know where the workers are, what they are doing, etc.

If you look at these applications carefully, all of them have the same kind of platform — they all need sensor data, predictive analytics, a connection to back-end systems, secure authentication and communication from devices to apps, etc. This list comprises the first four apps we have built. We are now taking the blockchain and combining it with IoT applications. **END** 